METHOD AND BASELINE

Benchmarking ANN search

SIFT-1M and GloVe-100 | HNSW, IVF-Flat, IVF-PQ | recall@10 against single-query throughput

METHOD

SIFT has **1,000,000** 128-D Euclidean vectors; GloVe has **1,183,514** 100-D vectors. We unit-normalize GloVe base and queries, then use squared Euclidean distance throughout. Original queries 1-5,000 are for tuning; all reported public-data results use held-out queries 5,001-10,000. Full-size sweeps chose build and search settings from 1,000 tuning queries; final searches use all base vectors and three timing repetitions.

FAISS **IndexFlatL2** supplies exact top ten. Our recall function counts distinct retrieved IDs whose true source-vector distance is no greater than the exact tenth-neighbour distance, including ties. Against all 10,000 stored references, both datasets give tie-aware recall 1.0 in both directions; only **65 SIFT** and **1 GloVe** top-ten ID sets differ, all at tie boundaries.

Machine: Intel Core Ultra 7 155H, 22 logical CPUs, 14.6 GiB RAM. Python 3.13.9, FAISS 1.15.1, NumPy 2.3.5, h5py 3.15.1. Timed search uses one FAISS/OpenMP thread and one query per call; full HNSW construction uses four threads. Build time includes training and insertion; index size is serialized FAISS bytes.

EXACT SEARCH: MEDIAN OVER 3 REPETITIONS

Dataset	Vectors	Single QPS	Batch QPS	Single ms/q
SIFT-1M	1,000,000	36.1	116.6	27.7
GloVe-100	1,183,514	44.8	116.6	22.3

Batch exact search reuses vector blocks and pays the Python-to-FAISS call overhead once. It is therefore faster than one-at-a-time exact search even with one thread; the ANN comparisons below use only the single-query baseline.

SIFT-1M TRADE-OFF

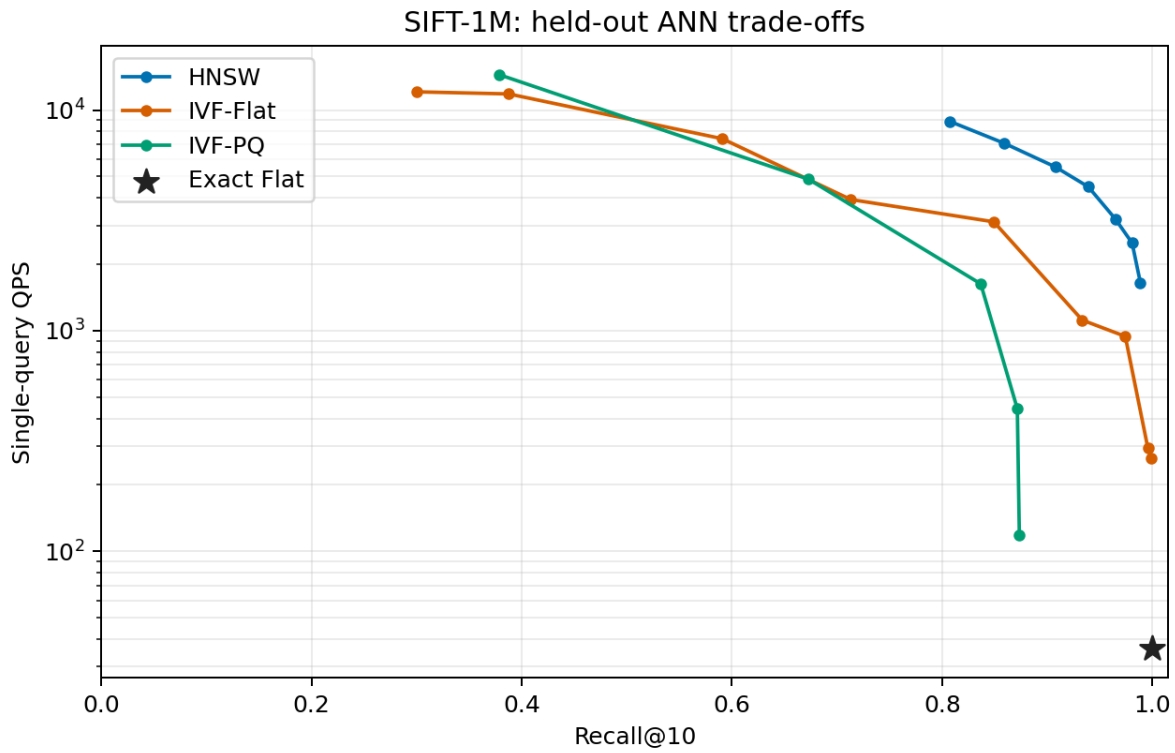


Figure 1. Held-out recall@10 versus single-query QPS; curves show measured Pareto points, with exact Flat at recall 1.0.

MAIN BENCHMARK

Recall, speed and index cost

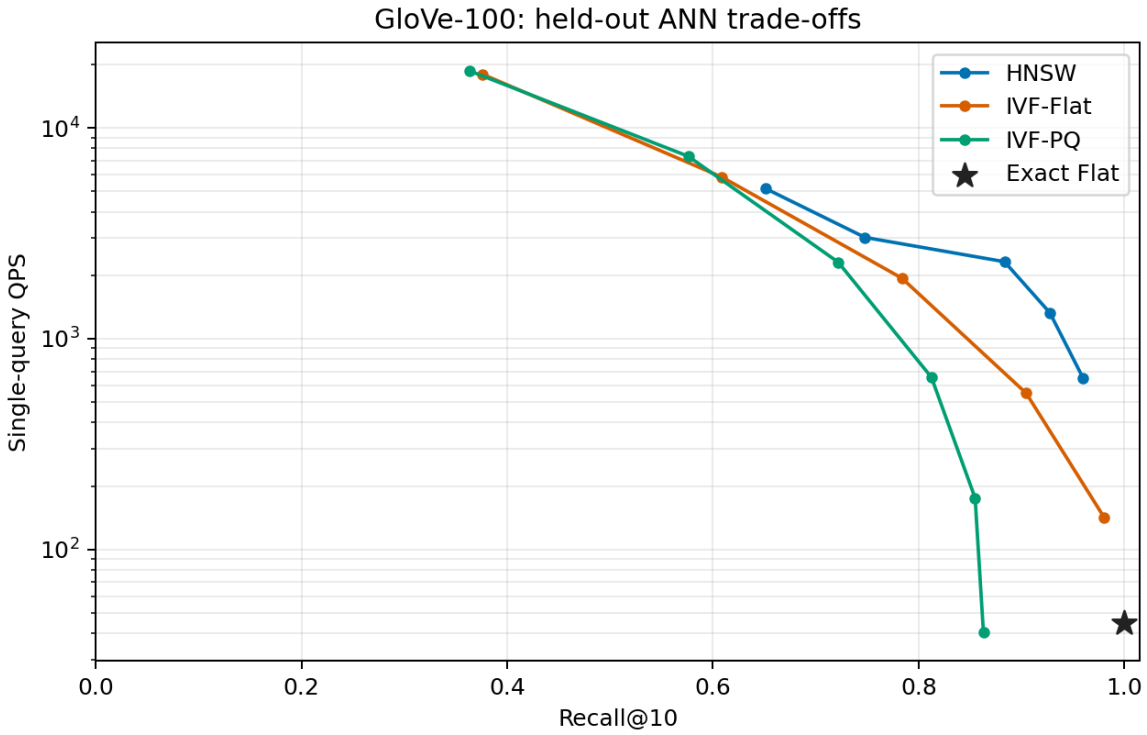


Figure 2. GloVe-100 held-out recall@10 versus single-query QPS; exact Flat is the reference point.

QPS AT FROZEN TUNING SELECTIONS

Index	SIFT @0.90	SIFT @0.95	GloVe @0.90	GloVe @0.95
HNSW	4,493	3,219	1,328	652
IVF-Flat	1,118	943	551	142
IVF-PQ	not reached	not reached	not reached	not reached
Exact Flat	36	36	45	45

A threshold entry reports the setting frozen on tuning data, not an interpolation. IVF-PQ reached at most **0.8735** on SIFT and **0.8630** on GloVe in the tested eight-bit grids; it missed even 0.90.

AT RECALL 0.95: BUILD TIME AND SERIALIZED SIZE

Dataset	Index	Recall	Build s	Size MiB	Bytes/vector
SIFT	HNSW	0.9647	208.6	625.9	656.3
SIFT	IVF-Flat	0.9743	158.8	497.9	522.1
SIFT	IVF-PQ	<0.95	--	--	--
GloVe	HNSW	0.9598	509.9	758.6	672.2
GloVe	IVF-Flat	0.9803	17.4	460.9	408.4
GloVe	IVF-PQ	<0.95	--	--	--

HNSW is fastest at 0.95 on both datasets: **3,219 versus 943 QPS** for SIFT and **652 versus 142 QPS** for GloVe (HNSW versus IVF-Flat). The tested IVF-PQ remained below target even with every coarse list probed; compressed distance ranking can still misorder candidates.

Which GloVe queries are hard?

Preregistered hypothesis (commit 82d9cdc): local contrast should make high-contrast queries easier for all three indexes, with the largest bin-4 minus bin-1 gain for IVF-PQ. For each held-out query, score = (d10 - d1) / d10 using true squared L2 distances to exact neighbours. Sort by score (original query row breaks ties) into four bins of 1,250; bin 1 has the smallest contrast.

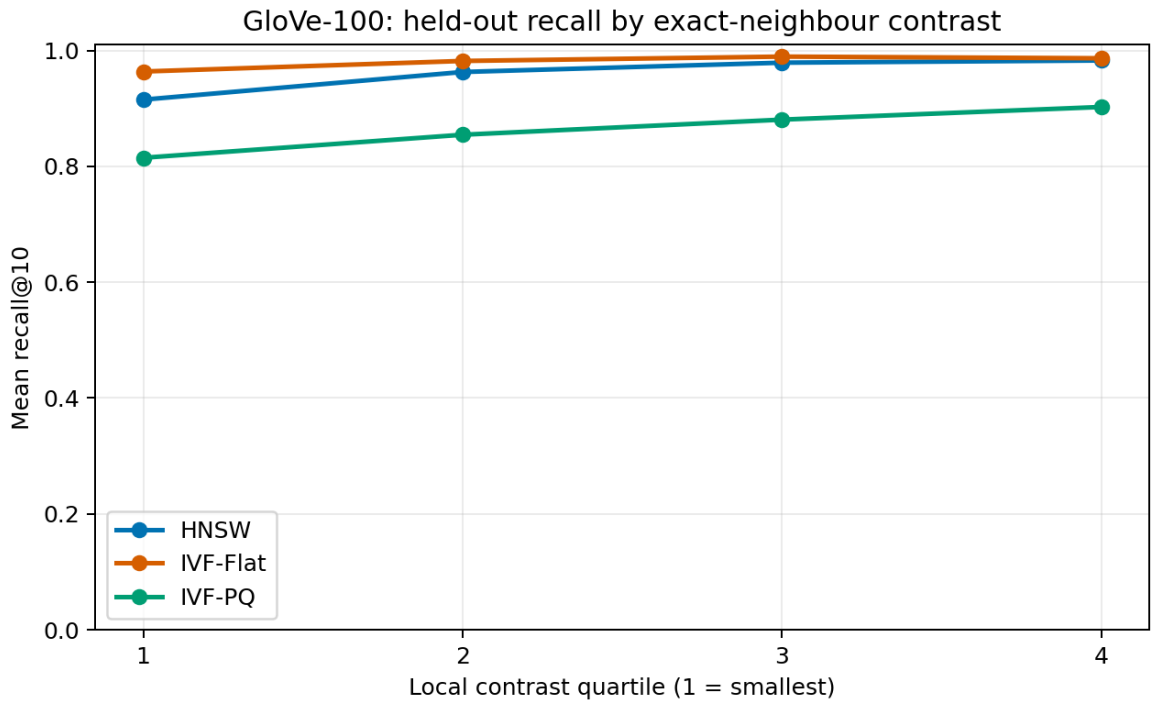


Figure 3. Recall@10 by exact-neighbour contrast, using frozen full-size GloVe settings.

HELD-OUT RECALL BY BIN

Bin	Contrast range	Count	HNSW	IVF-Flat	IVF-PQ
1	0.0262-0.1051	1,250	0.9150	0.9636	0.8147
2	0.1051-0.1660	1,250	0.9628	0.9818	0.8545
3	0.1660-0.2804	1,250	0.9789	0.9894	0.8806
4	0.2804-0.9185	1,250	0.9826	0.9865	0.9024

Outcome: all three improve from bin 1 to 4; endpoint gains are HNSW +0.0676, IVF-Flat +0.0229 and IVF-PQ +0.0877. IVF-Flat dips slightly from bin 3 to 4, so the monotonic prediction holds only for HNSW and IVF-PQ. Close neighbours may be displaced by approximate ranking, particularly by PQ distance error; probing all GloVe IVF-PQ lists still leaves recall at 0.8630.

The score measures spread within the top ten, not the tenth-to-eleventh gap, centroid-boundary position or graph reachability. It describes association, not a causal effect. HNSW and IVF-Flat preserve source vectors; PQ compresses them. The three settings have different search effort, so gap sizes are not controlled comparisons of intrinsic sensitivity.

When clusters become unbalanced

Preregistered hypothesis (commit 379f0d9): imbalance should reduce IVF-Flat recall more than HNSW, especially for rare-cluster queries. Seed 20261021 produces 96,000 32-D vectors around 16 fixed Gaussian centres and 1,024 separate, fixed queries (64 per cluster). Only the base population ratio changes: 1:1 control, then 4:1, 16:1 and 64:1. Base noise, insertion order, centre spread and index settings stay fixed.

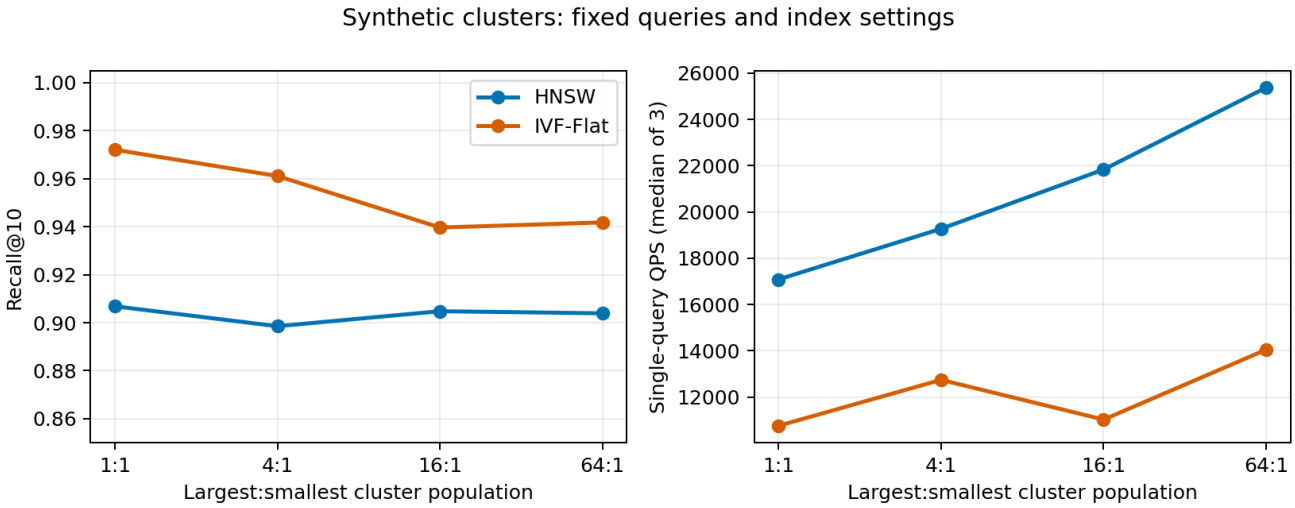


Figure 4. Exact-reference recall@10 and median one-query-at-a-time QPS; three repetitions of 20 passes at each ratio.

FIXED SEARCH SETTINGS AND OBSERVED RECALL

Ratio	Min-max base/cluster	HNSW recall	IVF-Flat recall	IVF-Flat QPS
1:1	6000-6000	0.9068	0.9721	10,754
4:1	2744-10977	0.8985	0.9611	12,740
16:1	1068-17089	0.9048	0.9396	11,022
64:1	368-23524	0.9039	0.9418	14,046

HNSW (M=16, efConstruction=100, efSearch=32) changes from 0.9068 to 0.9039; IVF-Flat (nlist=64, nprobe=4) falls from 0.9721 to 0.9418. The predicted **larger IVF-Flat decline** occurred, but the rare-cluster explanation failed: its four smallest clusters kept recall 1.0000, while the four largest fell from 0.9262 to 0.7672.

Assigning learned centroids to their nearest generating centre, the four largest clusters drew 22 then 45 of 64; fixed nprobe=4 covers less of a populous cluster even as mean scanned candidates rose from 5,625 to 6,644. This supports a partition-coverage explanation, without locating every missed neighbour. HNSW graph paths were not inspected. The seed reproduced all data hashes and recall on a second run; timing varies. Dimension, spread, query mix and search effort were not varied.

What the measurements imply

1. FASTEST AT RECALL 0.95

HNSW wins on both datasets, with the measured pairs above: SIFT 3,219 versus 943 QPS and GloVe 652 versus 142 QPS against IVF-Flat. Its graph search follows promising links instead of scanning the full base, while keeping exact vector distances. This is a comparison of frozen tuned settings on one machine, not a universal ranking.

2. BUILD AND MEMORY COST; WHAT PQ BUYS

At 0.95, HNSW needs 208.6 s and 656.3 bytes/vector on SIFT, and 509.9 s and 672.2 bytes/vector on GloVe. IVF-Flat needs 158.8 s / 522.1 bytes/vector and 17.4 s / 408.4 bytes/vector, respectively. The best tested IVF-PQ builds use only **72.7** and **58.4** bytes/vector (build 65.4 and 54.9 s), but top out at recall 0.8735 and 0.8630. PQ trades memory for quantized ranking error; scanning all coarse lists did not restore 0.90.

3. WHEN HEADLINE NUMBERS MISLEAD

GloVe bin-1 recall is below bin-4 recall for every index (for IVF-PQ, 0.8147 versus 0.9024), so an overall mean hides hard queries. In synthetic data, overall IVF-Flat recall fell only 0.0303 across imbalance, yet large-cluster recall fell 0.1590. Query mix can therefore conceal who loses accuracy; the rare-cluster prediction was wrong.

4. 150,000 OR 10 MILLION VECTORS; 100 MS AND 16 GB

150K: choose exact Flat when perfect recall matters. A linear extrapolation of measured one-at-a-time exact time gives about 4.2 ms/query (SIFT) or 2.8 ms (GloVe), well below 100 ms. **10M:** exact extrapolates to about 277 and 189 ms, beyond budget; start with frozen HNSW 0.95 settings (SIFT M=16, efConstruction=120, efSearch=64; GloVe M=32, efConstruction=160, efSearch=512). Serialized size projected linearly is roughly 6.6-6.7 GB, within 16 GB. Re-tune and measure at 10M: recall, latency and peak build RAM were not tested there.

5. TWO BENCHMARK LIMITATIONS AND WHETHER TESTED

Query distribution: a single overall recall can hide difficult queries or cluster-specific misses. *Tested* with GloVe contrast bins and fixed-mix synthetic clusters. **Scale and resource extrapolation:** index overhead, training quality, caches and build peak RAM may change from about 1M to 10M; linear projections can mislead. *Not tested* at 10M. Single-thread search also excludes concurrent serving, which was not tested.

Traceability. Main figures and tables map to run IDs in results/results.csv; Part 3 bin rows link to their final runs; Part 4 uses eight seeded settings with three timing rows each. Source and exact-neighbour hashes are recorded. Hypothesis commits predate experiment-result commits. All reported public-data settings were frozen from tuning queries.